

## **SOLUTION QUIZ-06 BSE-6A**

### **QUESTION-01**

1. B
2. B
3. B
4. C
5. B

### **QUESTION-02**

1. The source mac address at point 3 is 40-4C-C5-CF-15-F7
2. The destination mac address at point 3 is 03-AF-2C-4F-7F-4B
3. The source IP address at point 3 is 128.119.63.9
4. The destination IP address at point 3 is 128.119.124.52
5. Yes, the datagram is being passed through a router, which causes a subnet change.
6. The source mac address at point 4 is 66-50-CE-6A-7B-FF
7. The destination mac address at point 4 is 3E-78-A3-58-96-91
8. No, datagrams can be sent across the subnet via the link layer in one go.

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|-----------------------------------------------------------------------------------|---------------------|----------------------------------|---------------------|--------------------|
|  | <b>Course Name:</b> | <b>Computer Networks</b>         | <b>Course Code:</b> | <b>CS 3001</b>     |
|                                                                                   | <b>Program:</b>     | <b>BS (Software Engineering)</b> | <b>Semester:</b>    | <b>Spring 2026</b> |
|                                                                                   | <b>Duration:</b>    | <b>20 minutes</b>                | <b>Total Marks:</b> | <b>15</b>          |
|                                                                                   | <b>Paper Date:</b>  | <b>05-May-2026</b>               | <b>Section</b>      | <b>6A , 6B</b>     |
|                                                                                   | <b>Exam Type:</b>   | <b>Quiz 6 - Chapter 6</b>        | <b>Page(s):</b>     | <b>2</b>           |

**Student Name**

**Roll No.**

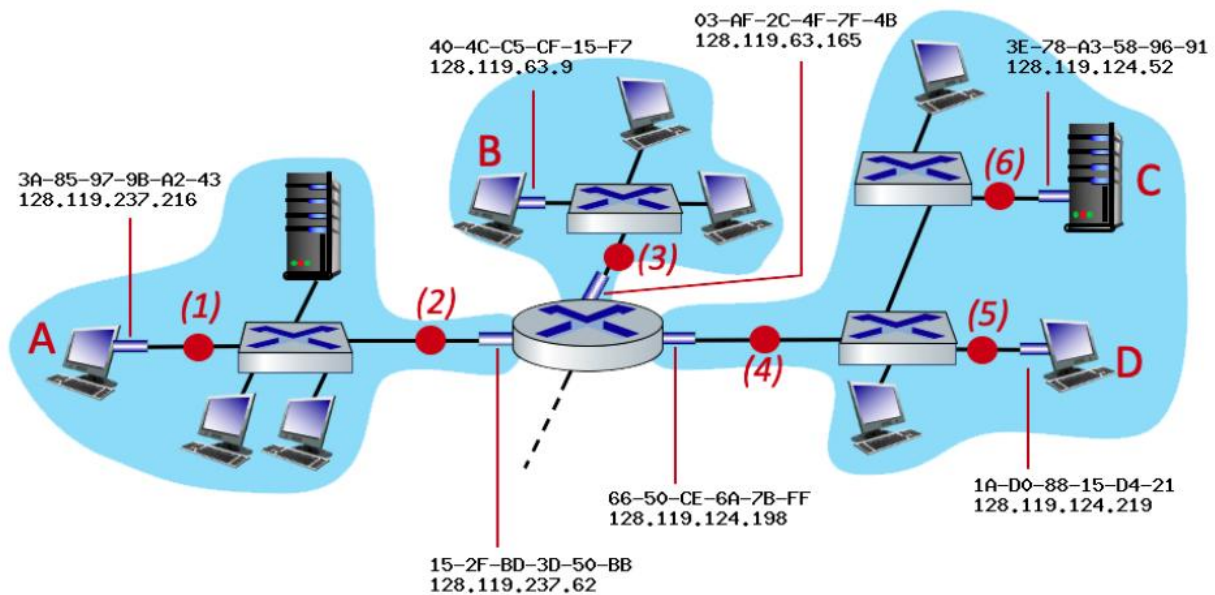
**Section:**

**Q1. Encircle the correct option:**

**[1\*5 = 5 Marks] [CLO 3]**

1. Host A sends a frame to Host B via Router R. Which field(s) in the layer 3 and/or layer 2 header/trailer change as it moves from Host A -> Router link -> Host B link?
  - a. Source IP and Destination IP.
  - b. Source MAC, Destination MAC and TTL.
  - c. Source MAC and Destination MAC only.
  - d. The CRC checksum only.
2. A student powers on a laptop in a coffee shop. The laptop has no static IP and the cache is empty. Which of the following is the CORRECT chronological order of the very first messages sent by the laptop?
  - a. ARP Request -> DNS Query -> DHCP Discover
  - b. DHCP Discover -> ARP Request -> DNS Query
  - c. DHCP Discover -> DNS Query -> ARP Request
  - d. SYN -> DHCP Discover -> HTTP GET
3. Consider a generator polynomial  $G(x)$  with the binary representation 101101 (6 bits). What is the maximum burst error length for which detection is guaranteed?
  - a. 4 bits
  - b. 5 bits
  - c. 6 bits
  - d. 3 bits
4. You are analyzing a generator polynomial  $G(x) = x^3 + x^2 + x$  (Binary: 1110). Which of the following statements accurately describes the weaknesses of this generator?
  - a. It can detect single-bit errors, but fails to detect odd-numbered errors.
  - b. It can detect odd-numbered errors, but fails to detect single-bit errors.
  - c. It fails to detect both single-bit errors AND fails to guarantee detection of all odd-numbered errors.
  - d. It is a robust generator that detects both single-bit and odd-numbered errors effectively.
5. A 2-D parity check can:
  - a. Detect a single bit error
  - b. Detect & correct a single bit error
  - c. Detect multiple bits errors
  - d. Detect & correct multiple bits errors

**Q2:** Consider the figure below. The IP and MAC addresses are shown for nodes A, B, C and D, as well as for the router's interfaces. **[10 Marks] [CLO 3]**



Consider an IP datagram being sent from node B to node C. Please Answer the following:

1. What is the source mac address at point 3?  
Answer: \_\_\_\_\_
2. What is the destination mac address at point 3?  
Answer: \_\_\_\_\_
3. What is the source IP address at point 3?  
Answer: \_\_\_\_\_
4. What is the destination IP address at point 3?  
Answer: \_\_\_\_\_
5. Do the source and destination mac addresses change at point 4? Answer with yes or no. **[2]**  
Answer: \_\_\_\_\_
6. What is the source mac address at point 4?  
Answer: \_\_\_\_\_
7. What is the destination mac address at point 4?  
Answer: \_\_\_\_\_
8. Do the source and destination mac addresses change at point 6? Answer with yes or no. **[2]**  
Answer: \_\_\_\_\_

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## Quiz 6: Link Layer (Chapter 6)

Name: \_\_\_\_\_ Roll No: \_\_\_\_\_ Section: BSE-6B1 (Spring 2026)

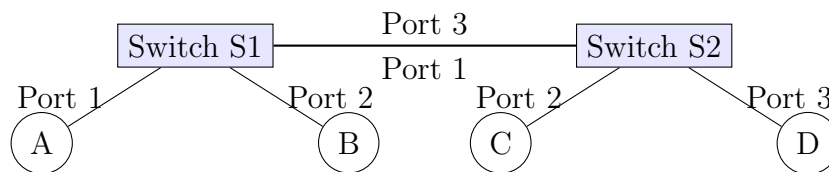
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### 1. (5 points) Switch Tables

Consider the switched LAN topology shown below. The switch tables are initially empty. The following sequence of frame transmissions occurs:

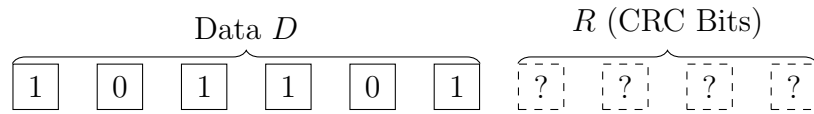
1. Node A sends a frame to Node D.
2. Node D replies with a frame to Node A.
3. Node C sends a frame to Node D.

Show the state of the Switching Table for **Switch S2** after these three events. Format: (MAC Address, Interface). Also, show the same for **Switch S1**.



2. (10 points) **Cyclic Redundancy Check (CRC) and Burst Errors**

Consider a data transmission scenario using a CRC generator polynomial  $G(x) = x^4 + x + 1$ . The data string to be transmitted is  $D = 101101$ .



1. Analytically calculate the 4-bit CRC remainder  $R$ . Show the long division in binary modulo-2 arithmetic. What is the actual bit-string transmitted by sender?
2. A "burst error" of length  $k$  is a contiguous sequence of bits in which the first and last bits are errors, and the intermediate bits may or may not be errors. Can this specific generator  $G(x)$  can detect **all** burst errors of length  $k = 3$ . Give reason?
3. Suppose a burst error occurs during transmission such that the received bit string has the 3rd and 4th bits (counting from the left, 1-based index) inverted. Does the receiver accept or reject this frame?

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## Quiz 6: Link Layer (Chapter 6)

Name: \_\_\_\_\_ Roll No: \_\_\_\_\_ Section: BSE-6B1 (Spring 2026)

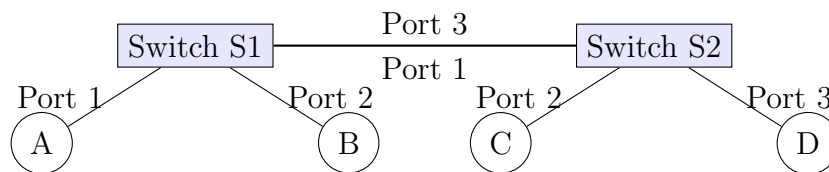
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### 1. (5 points) Switch Tables

Consider the switched LAN topology shown below. The switch tables are initially empty. The following sequence of frame transmissions occurs:

1. Node A sends a frame to Node D.
2. Node D replies with a frame to Node A.
3. Node C sends a frame to Node D.

Show the state of the Switching Table for **\*\*Switch S2\*\*** after these three events. Format: (MAC Address, Interface). Also, show the same for **\*\*Switch S1\*\***.



### Solution:

#### • Switch S1 Table:

| MAC Address | Interface |
|-------------|-----------|
| A           | Port 1    |
| D           | Port 3    |

*Note: S1 learns A from step 1. S1 learns D from step 2. S1 never sees the frame from C (step 3) because S2 knows D is on Port 3 and does not flood the frame back to S1.*

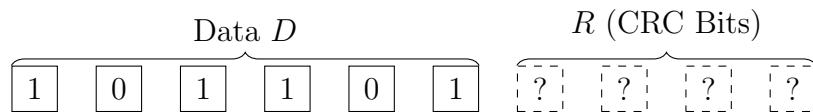
#### • Switch S2 Table:

| MAC Address | Interface |
|-------------|-----------|
| A           | Port 1    |
| D           | Port 3    |
| C           | Port 2    |

*Note: S2 learns A from step 1 (incoming on Port 1). S2 learns D from step 2 (incoming on Port 3). S2 learns C from step 3 (incoming on Port 2).*

## 2. (10 points) Cyclic Redundancy Check (CRC) and Burst Errors

Consider a data transmission scenario using a CRC generator polynomial  $G(x) = x^4 + x + 1$ . The data string to be transmitted is  $D = 101101$ .



1. Analytically calculate the 4-bit CRC remainder  $R$ . Show the long division in binary modulo-2 arithmetic. What is the actual bit-string transmitted by sender?
2. A "burst error" of length  $k$  is a contiguous sequence of bits in which the first and last bits are errors, and the intermediate bits may or may not be errors. Can this specific generator  $G(x)$  can detect **all** burst errors of length  $k = 3$ . Give reason?
3. Suppose a burst error occurs during transmission such that the received bit string has the 3rd and 4th bits (counting from the left, 1-based index) inverted. Does the receiver accept or reject this frame?

### Solution:

#### 1. CRC Calculation:

- Generator  $G(x) = x^4 + x + 1 \Rightarrow$  **10011** (5 bits).
- Data  $D = 101101$ . Append 4 zeros: **1011010000**.
- Perform Binary Division (XOR):

```

      101101
10011 ) 1011010000
      10011
      ----
        01011
        00000
        ----
          10110
          10011
          ----
            01010
            00000
            ----
              10100
              10011
              ----
                01110
                00000
                ----
                  1110  <-- Remainder (R)
  
```

The remainder  $R$  is **1110**.

The transmitted string is **1011011110**.

#### 2. Burst Error Detection:

Yes. A generator polynomial of degree  $r$  (here  $r = 4$ ) detects **all** burst errors of length  $k \leq r$ . Since the burst length  $k = 3$  and  $3 \leq 4$ , this error is guaranteed to be detected.

### 3. Receiver Check:

The 3rd and 4th bits are inverted.

Original Transmitted: 10**11**011110

Received Frame: 10**00**011110

Performing the division of 1000011110 by 10011:

$1000011110 \div 10011$  yields a remainder of **1011**.

Since the remainder is non-zero ( $\neq 0000$ ), the receiver **rejects** the frame.

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|                                                                                   | <b>Program:</b>     | <b>BS (Computer Science)</b> | <b>Semester:</b>    | <b>Spring 2026</b> |
|                                                                                   | <b>Duration:</b>    | <b>15 minutes</b>            | <b>Total Marks:</b> | <b>15</b>          |
|                                                                                   | <b>Paper Date:</b>  |                              | <b>Section</b>      | <b>6B1</b>         |
|                                                                                   | <b>Exam Type:</b>   | <b>Quiz 6 - Chapter 6</b>    | <b>Page(s):</b>     | <b>2</b>           |

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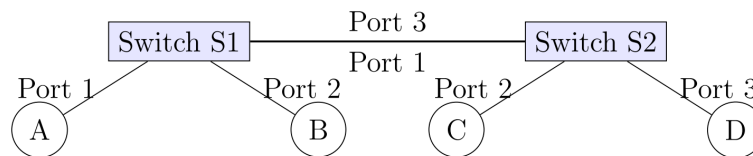
**Q1. Carefully read the Question before attempting:**

**[5 marks]**

Consider the switched LAN topology shown below. The switch tables are initially empty. The following sequence of frame transmissions occurs:

1. Node A sends a frame to Node D.
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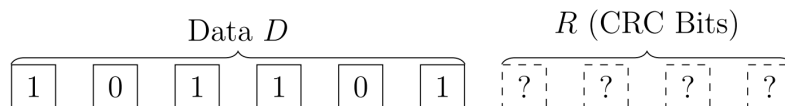
Show the state of the Switching Table for **Switch S2** after these three events. Format: (MAC Address, Interface). Also, show the same for **Switch S1**.



**Q2. Carefully read the Question before attempting:**

**[10 marks]**

Consider a data transmission scenario using a CRC generator polynomial  $G(x) = x^4 + x + 1$ . The data string to be transmitted is  $D = 101101$ .



1. Analytically calculate the 4-bit CRC remainder  $R$ . Show the long division in binary modulo-2 arithmetic. What is the actual bit-string transmitted by the sender?
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|                                                                                   | <b>Paper Date:</b>  |                              | <b>Section</b>      | <b>6B1</b>         |
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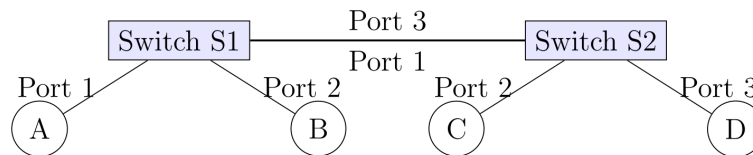
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**Solution:**

• **Switch S1 Table:**

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|-------------|-----------|
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*Note: S1 learns A from step 1. S1 learns D from step 2. S1 never sees the frame from C (step 3) because S2 knows D is on Port 3 and does not flood the frame back to S1.*

• **Switch S2 Table:**

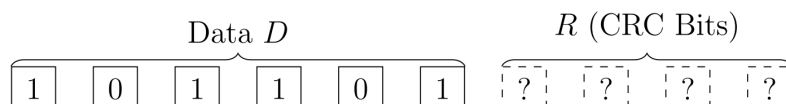
| MAC Address | Interface |
|-------------|-----------|
| A           | Port 1    |
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| C           | Port 2    |

*Note: S2 learns A from step 1 (incoming on Port 1). S2 learns D from step 2 (incoming on Port 3). S2 learns C from step 3 (incoming on Port 2).*

**Q2. Carefully read the Question before attempting:**

**[10 marks]**

Consider a data transmission scenario using a CRC generator polynomial  $G(x) = x^4 + x + 1$ . The data string to be transmitted is  $D = 101101$ .



1. Analytically calculate the 4-bit CRC remainder  $R$ . Show the long division in binary modulo-2 arithmetic. What is the actual bit-string transmitted by the sender?
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3. Suppose a burst error occurs during transmission such that the received bit string has the 3rd and 4th bits (counting from the left, 1-based index) inverted. Does the receiver accept or reject this frame?

### Solution:

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- Generator  $G(x) = x^4 + x + 1 \Rightarrow 10011$  (5 bits).
- Data  $D = 101101$ . Append 4 zeros: 1011010000.
- Perform Binary Division (XOR):

```

      101101
10011 ) 1011010000
      10011
      ----
        01011
        00000
        ----
          10110
          10011
          ----
            01010
            00000
            ----
              10100
              10011
              ----
                01110
                00000
                ----
                  1110  <-- Remainder (R)

```

The remainder  $R$  is **1110**.

The transmitted string is 101101**1110**.

#### 2. Burst Error Detection:

Yes. A generator polynomial of degree  $r$  (here  $r = 4$ ) detects **all** burst errors of length  $k \leq r$ . Since the burst length  $k = 3$  and  $3 \leq 4$ , this error is guaranteed to be detected.

#### 3. Receiver Check:

The 3rd and 4th bits are inverted.

Original Transmitted: 10**11**011110

Received Frame: 10**00**011110

Performing the division of 1000011110 by 10011:

$1000011110 \div 10011$  yields a remainder of **1011**.

Since the remainder is non-zero ( $\neq 0000$ ), the receiver **rejects** the frame.

|                                                                                   |              |                       |              |           |
|-----------------------------------------------------------------------------------|--------------|-----------------------|--------------|-----------|
|  | Course Name: | Computer Networks     | Course Code: | CS 3001   |
|                                                                                   | Program:     | BS (Computer Science) | Semester:    | Fall 2023 |
|                                                                                   | Duration:    | 20 minutes            | Total Marks: | 15        |
|                                                                                   | Paper Date:  | 05-December-2023      | Section      | 5J        |
|                                                                                   | Exam Type:   | Quiz 6 - Chapter 6    | Page(s):     | 2         |

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Section:

**Q1. Encircle the correct option:**

[5 marks] [CLO 3]

1. Pure (Un-slotted) ALOHA is:
  - a) Highly efficient
  - b) Highly centralized
  - c) Highly decentralized
  - d) Fully decentralized
2. What is the length of preamble in frame header?
  - a) 4 bytes
  - b) 5 bytes
  - c) 6 bytes
  - d) 7 bytes
3. It is possible to detect and correct multiple bit errors on link layer using:
  - a) CRC
  - b) Ethernet
  - c) 2 - D parity scheme
  - d) None of the above

**True/False:**

1. The data link layer divides the stream of bits received from the network layer into data units called Datagram. [T / F]
2. Flow control can be implemented at datalink layer. [T / F]

**Q2.** X wants to send a data string  $D = 11000$  to Y using the CRC generator string G equivalent to generator polynomial  $G(x) = x^3 + x + 1$ . What is the value of check bit sequence? Show complete working. **[10 Marks] [CLO 3]**

## Solution Quiz#6(5J)

### MCQ's

1. d
2. d
3. d

### T/F

1. F
2. T

### Q2. Solution:-

Message = 11000

CRC polynomial =  $x^3+x+1$

CRC sequence = 100

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|                                                                                   | <b>Paper Date:</b>  | <b>05-December-2023</b>      | <b>Section</b>      | <b>5K</b>        |
|                                                                                   | <b>Exam Type:</b>   | <b>Quiz 6 - Chapter 6</b>    | <b>Page(s):</b>     | <b>3</b>         |

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**Q1. Encircle the correct option:**

**[5 marks] [CLO 3]**

- Which of the following is the multiple access protocol for wired network channel?
  - CSMA/CD
  - CSMA/CA
  - Both
  - All
- The data link layer takes the packet from\_\_\_\_\_and encapsulates them into frame for transmission.
  - Segment
  - Frames
  - Datagram
  - Message
- It is possible to detect and correct multiple bit errors on link layer using:
  - CRC
  - Ethernet
  - 2 - D parity scheme
  - None of the above

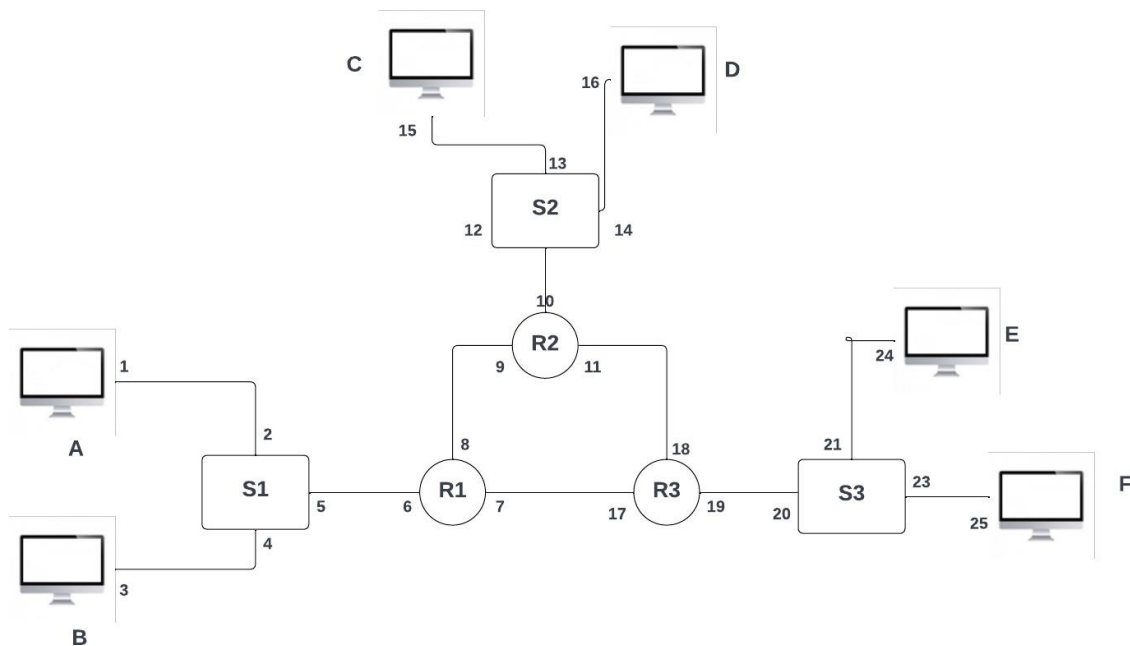
**True/False:**

- Network Layer lies below the data link layer. [ T / F ]
- Minimum payload length in frame header is 56 bytes. [ T / F ]

**Q2.** Consider the network scenario given in Figure comprising six end-systems/hosts: A, B, C, D, E and F with MAC addresses: MAC-A, MAC-B, MAC-C, MAC-D, MAC-E, MAC-F and IP addresses: IP-A, IP-B, IP-C, IP-D, IP-E, IP-F respectively. DHCP is enabled for the given network topology. S1, S2 and S3 denote the switches while routers are represented by R1, R2 and R3. Connecting lines depict the network connections (wired) between all the network nodes with individual digits on each end representing the interface number of the connected node.

Furthermore, for IP and MAC addresses of the interfaces, IP-(Interface number) and MAC-(Interface number) format is in use. For example, IP-6 and MAC-6 refers to the IP and MAC addresses of Router1's interface 6.

**[4 + 6 Marks] [CLO 3]**



**Part (a):** Consider the following ARP table for interface 10 of the router R2 and complete the table with the help of entries given. For each entry, please give a brief reason for your choice. **[4 Marks]**

| <u>IP Address</u> | <u>Mac Address</u> | <u>Short Reason</u> |
|-------------------|--------------------|---------------------|
| IP-15             |                    |                     |
|                   | MAC-D              |                     |

**Part(b):** Consider the following ARP table for interface 19 of the router R3 and specify in the entries given are correct or not. For each entry, please give a brief reason for your choice.  
**[6 Marks]**

| <u><b>IP Address</b></u> | <u><b>Mac Address</b></u> | <u><b>Correct Entry?(yes or no)</b></u> | <u><b>Short Reason</b></u> |
|--------------------------|---------------------------|-----------------------------------------|----------------------------|
| IP-A                     | MAC-A                     |                                         |                            |
| IP-20                    | MAC-20                    |                                         |                            |
| IP-E                     | MAC-24                    |                                         |                            |

## Solution Quiz#6(5K)

### MCQ's

1. a
2. c
3. d

### T/F

1. F
2. F

### Q2.

#### Part-A:

| <u>IP Address</u> | <u>Mac Address</u> | <u>Short Reason</u> |
|-------------------|--------------------|---------------------|
| IP-15             | MAC-C/MAC-15       | Same subnet         |
| IP-D/IP16         | MAC-D              | Same subnet         |

#### Part-B:

| <u>IP Address</u> | <u>Mac Address</u> | <u>Correct Entry?(yes or no)</u> | <u>Short Reason</u>    |
|-------------------|--------------------|----------------------------------|------------------------|
| IP-A              | MAC-A              | No                               | Not on the same subnet |
| IP-20             | MAC-20             | No                               | Switch interface       |
| IP-E              | MAC-E              | Yes                              | Same subnet            |

## SOLUTION QUIZ-06 BCS-5A

### QUESTION-01

1. A
2. D
3. A

### TRUE/FALSE

- i. True
- ii. True

### QUESTION-02

1. Host F
2. Router C
3. Source: IP = IP<sub>B</sub>, MAC = MAC<sub>B</sub> ; Destination: IP = IP<sub>A</sub>, MAC = MAC<sub>C1</sub>
4. Source: IP = IP<sub>A</sub>, MAC = MAC<sub>A</sub> ; Destination: IP = IP<sub>F</sub>, MAC = MAC<sub>F</sub>

### QUESTION-03

1. Source MAC: AA-BB-CC-11-22-33, Destination MAC: FF-FF-FF-FF-FF-FF
2. Source MAC: CC-DD-EE-11-22-33, Destination MAC: AA-BB-CC-11-22-33

### QUESTION-04

**a)** Packet on link from S1 to R1:

Source IP: IP1  
Destination IP: IP12

Source MAC: MAC1  
Destination MAC: MAC4

**b)** Packet on link from NAT to R1:

Source IP: IP12  
Destination IP: IP1

Source MAC: MAC6  
Destination MAC: MAC5

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|                                                                                   | <b>Exam Type:</b>   | <b>Quiz 6 - Chapter 6</b>    | <b>Page(s):</b>     | <b>3</b>         |

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**Section:**

**Q1. Encircle the correct option:**

**[5 marks] [CLO 1]**

1. What is the purpose of ARP:
  - a) To map IP addresses to MAC addresses
  - b) To map MAC addresses to IP addresses
  - c) To map domain names to IP addresses
  - d) To map subnet masks to IP addresses
2. Pure (Un-slotted) ALOHA is:
  - a) Highly Efficient
  - b) Highly Centralized
  - c) Highly Decentralized
  - d) Fully Decentralized
3. In which network technology is CSMA/CD commonly used?
  - a) Ethernet
  - b) Wifi
  - c) Fiber Optics
  - d) Bluetooth

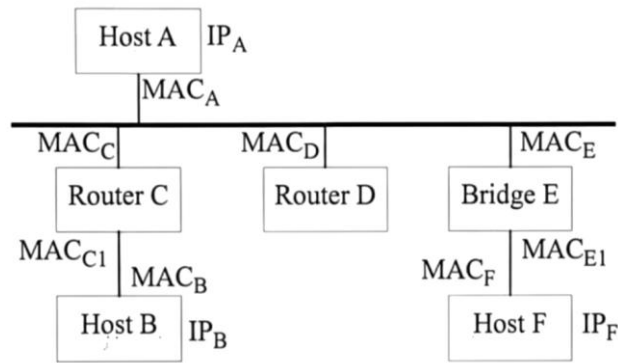
**True/False:**

1. In networks using CSMA/CD, there is a chance of collision. [T / F]
2. When an IP router between two Ethernet segments forwards an IP packet, it **does not** modify the destination IP address. [T / F]

**Q2:**

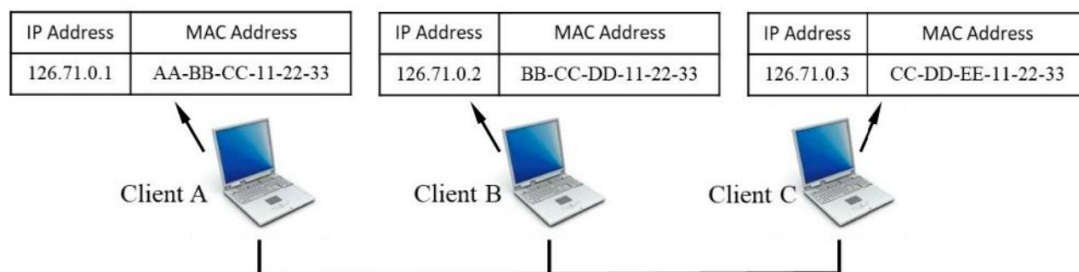
**[4 Marks] [CLO 1]**

In the Network shown below, forwarding tables of Routers are already populated by running a routing protocol, and forwarding table of bridge is already populated with self-learning:



- (1) Host A broadcasts an ARP Request frame to resolve the MAC Address of IP Address IP<sub>F</sub>. Which device (host, bridge or router) will respond with an ARP reply frame?
- (2) Host B broadcasts an ARP Request frame to resolve the MAC Address of IP Address IP<sub>F</sub>. Which device (host, bridge or router) will respond with an ARP reply frame?
- (3) Host B wants to send an IP Packet to Host A. What are the source and destination IP Address in the packet and the source and destination MAC address in the frame?
- (4) Host A wants to send an IP Packet to Host F. What are the source and destination IP Address in the packet and source and destination MAC address in the frame?

**Q3:** [1+1 = 2 Marks] [CLO 1]  
 Address resolution protocol (ARP) and Network address translation (NAT) are important in the information technology field:

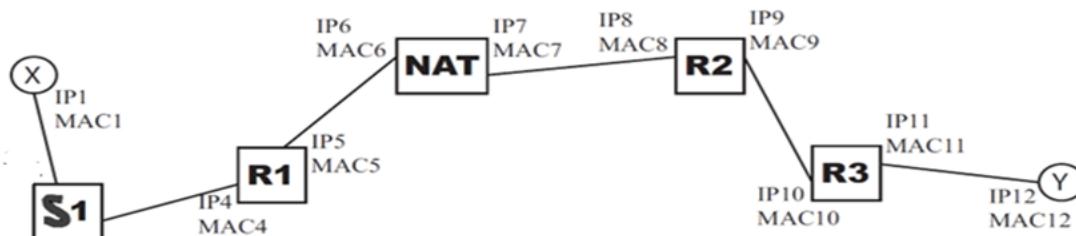


Suppose Client A knows that the IP Address of Client C is 126.71.0.3, but does not know the MAC Address of Client C. In this case, Client A can apply ARP to obtain the MAC Address of Client C.

1. First Client A will send a Frame to ask the MAC Address of the device with IP Address 126.71.0.3. Write down the source MAC address and the destination MAC address of this Frame:

2. Then Client C will send a frame to inform Client A of its MAC Address. Write down the source MAC Address and the Destination MAC Address of this frame:

**Q4:** A network topology is shown in the figure below. X and Y are hosts. The NAT IP6 is on the “inside” (i.e. private side) while IP7 is a global IP address. S1 is a Layer 2 switch, while R1, R2 and R3 are routers. Assume that X and Y are exchanging application layer traffic that uses TCP underneath. We are interested in source and destination IP and MAC on various packets in the network for this traffic. For packets on each of the following directional links, indicate all the four fields: **[2 + 2 = 4 Marks] [CLO 1]**



- a. For Packet on link from S1 to R1:

Source IP: \_\_\_\_\_  
Destination IP: \_\_\_\_\_

Source MAC: \_\_\_\_\_  
Destination MAC: \_\_\_\_\_

- b. Packet on link from NAT to R1.

Source IP: \_\_\_\_\_  
Destination IP: \_\_\_\_\_

Source MAC: \_\_\_\_\_  
Destination MAC: \_\_\_\_\_

## SOLUTION QUIZ-06 BCS-5C

### QUESTION-01

1. B
2. A
3. A

### TRUE/FALSE

- i. False
- ii. True

### QUESTION-02

Packet 01:

| <i>Header Type</i> | <i>Source</i> | <i>Destination</i> |
|--------------------|---------------|--------------------|
| Ethernet           | MAC 16        | MAC 15             |
| IP                 | 3.0.1.2       | 1.2.3.4            |

Packet 02:

| <i>Header Type</i> | <i>Source</i> | <i>Destination</i> |
|--------------------|---------------|--------------------|
| Ethernet           | MAC 1         | MAC 2              |
| IP                 | 10.0.0.2      | 3.0.1.2            |
|                    |               |                    |
|                    |               |                    |
|                    |               |                    |

Packet 03:

| <i>Header Type</i> | <i>Source</i> | <i>Destination</i> |
|--------------------|---------------|--------------------|
| Ethernet           | MAC 3         | MAC 4              |
| IP                 | 1.2.3.4       | 3.0.1.2            |
|                    |               |                    |
|                    |               |                    |
|                    |               |                    |

Packet 04:

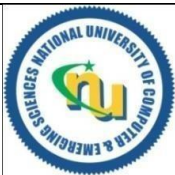
| <i>Header Type</i> | <i>Source</i> | <i>Destination</i> |
|--------------------|---------------|--------------------|
| Ethernet           | MAC 9         | MAC 10             |
| IP                 | 1.2.3.8       | 1.2.4.3            |
| IP                 | 1.2.3.6       | 2.0.0.1            |
| IP                 | 1.2.3.4       | 3.0.1.2            |
|                    |               |                    |

## QUESTION-02

1. Host H1 and H2
- 2.

| <b>Ethernet Source</b> | <b>Ethernet Destination</b> | <b>IP Source</b> | <b>IP Destination</b> |
|------------------------|-----------------------------|------------------|-----------------------|
| 02:ab:12:34:56:78      | ff:ff:ff:ff:ff:ff           | 192.168.1.1      | 192.168.1.2           |

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|                                                                                   | <b>Program:</b>     | <b>BS (Computer Science)</b> | <b>Semester:</b>    | <b>Fall 2025</b> |
|                                                                                   | <b>Duration:</b>    | <b>15 minutes</b>            | <b>Total Marks:</b> | <b>15</b>        |
|                                                                                   | <b>Paper Date:</b>  | <b>27-November-2025</b>      | <b>Section</b>      | <b>5C , 5A</b>   |
|                                                                                   | <b>Exam Type:</b>   | <b>Quiz 6 - Chapter 6</b>    | <b>Page(s):</b>     | <b>3</b>         |

**Student Name**

**Roll No.**

**Section:**

**Q1. Encircle the correct option:**

**[5 marks] [CLO 1]**

- In a switch, what information does the switch table store?
  - IP addresses of connected devices
  - Port and corresponding MAC addresses
  - Domain Name System (DNS) entries
  - Switch configuration settings
- CRC is used for:
  - Error Detection
  - Error Correction
  - Both of the above
  - None of the above
- In ALOHA, what happens if a collision occurs?
  - The data is retransmitted after a random time
  - The data is discarded
  - The sender waits for an acknowledgment
  - The sender reduces the transmission rate

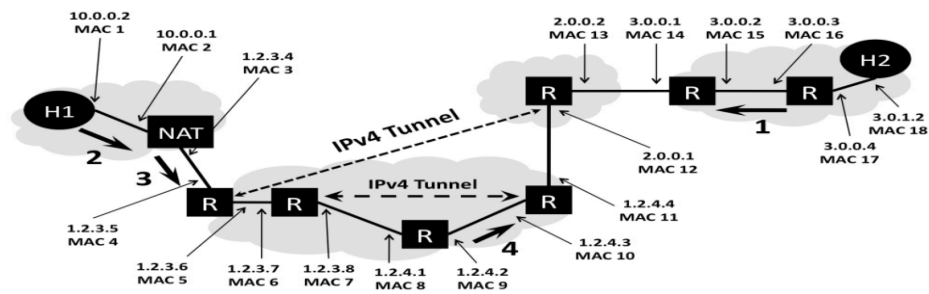
**True/False:**

- Pure (Un-slotted) Aloha is highly centralized. [T / F]
- In networks using CSMA/CD, there is a chance of collision. [T / F]

**Q2:**

**[6 Marks] [CLO 1]**

The figure below shows a network topology. The LAN on the left uses a NAT to connect to the Internet and includes a client host H1. The LAN on the right includes a webserver H2. Packets between the two endpoints are routed along the path shown by a heavy dark line, which includes two IPv4 tunnels. All packets which traverse the path use both tunnels. The various network interfaces have IP and MAC addresses as shown:



H1 has established an HTTP session with web server H2 and data packets are flowing between the two machines. You have to fill in the missing entries in the tables below, i.e. the source and destination address for the corresponding network and datalink layers for packets 2, 3, and 4 (these packets are all traveling from the client H1 to the server H2, as marked on the figure with heavy black arrows and numbers).

| Packet 1   |        |             |
|------------|--------|-------------|
| Header     | Source | Destination |
| Ethernet   | MAC 16 | MAC 15      |
| IP Address |        |             |

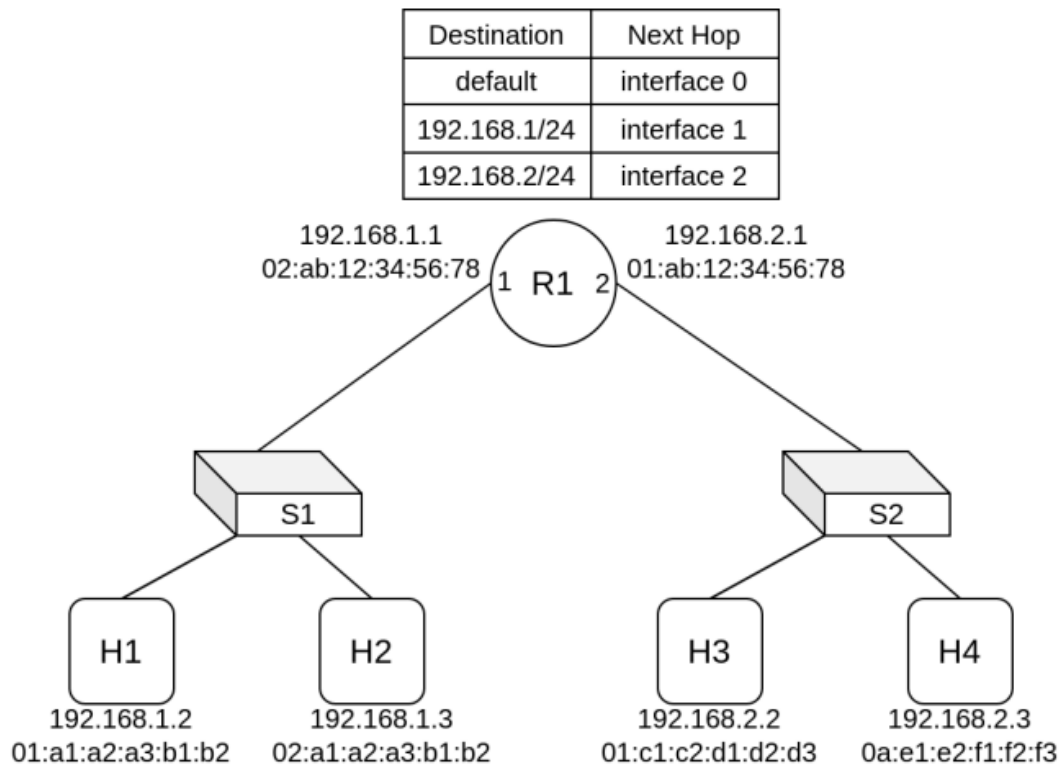
| Packet 2   |        |             |
|------------|--------|-------------|
| Header     | Source | Destination |
| Ethernet   |        |             |
| IP Address |        |             |

| Packet 3   |         |             |
|------------|---------|-------------|
| Header     | Source  | Destination |
| Ethernet   |         |             |
| IP Address | 1.2.3.4 | 3.0.1.2     |

| Packet 4   |        |             |
|------------|--------|-------------|
| Header     | Source | Destination |
| Ethernet   |        |             |
| IP Address |        |             |

**Q3:**

Consider the network topology shown in Figure below comprising router, switches, and hosts. The IP addresses and MAC addresses assigned to various interfaces are shown in the figure. Assume that the router's ARP table is empty: [2 + 2 = 4 Marks] [CLO 1]



Suppose that the following IP Packet's arrive at **interface 2** of Router:

| Ethernet Source   | Ethernet Destination | IP Source   | IP Destination |
|-------------------|----------------------|-------------|----------------|
| 0a:e1:e2:f1:f2:f3 | 01:ab:12:34:56:78    | 192.168.2.3 | 192.168.1.2    |

1. To successfully deliver the above IP Packet, the router needs to send an ARP Request. Which host(s) receive the Request?
2. Mention the following header fields for the ARP request packet sent by the router?
  - Ethernet Source:
  - Ethernet Destination:
  - IP Source:
  - IP Destination:

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|                                                                                   | <b>Program:</b>     | <b>BS (Computer Science)</b> | <b>Semester:</b>    | <b>Spring 2024</b> |
|                                                                                   | <b>Duration:</b>    | <b>15 minutes</b>            | <b>Total Marks:</b> | <b>15</b>          |
|                                                                                   | <b>Paper Date:</b>  | <b>02-May-2024</b>           | <b>Section</b>      | <b>6A</b>          |
|                                                                                   | <b>Exam Type:</b>   | <b>Quiz 6 - Chapter 6</b>    | <b>Page(s):</b>     | <b>3</b>           |

**Student Name**

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**Section:**

**Q1. Encircle the correct option:**

**[5 marks] [CLO 3]**

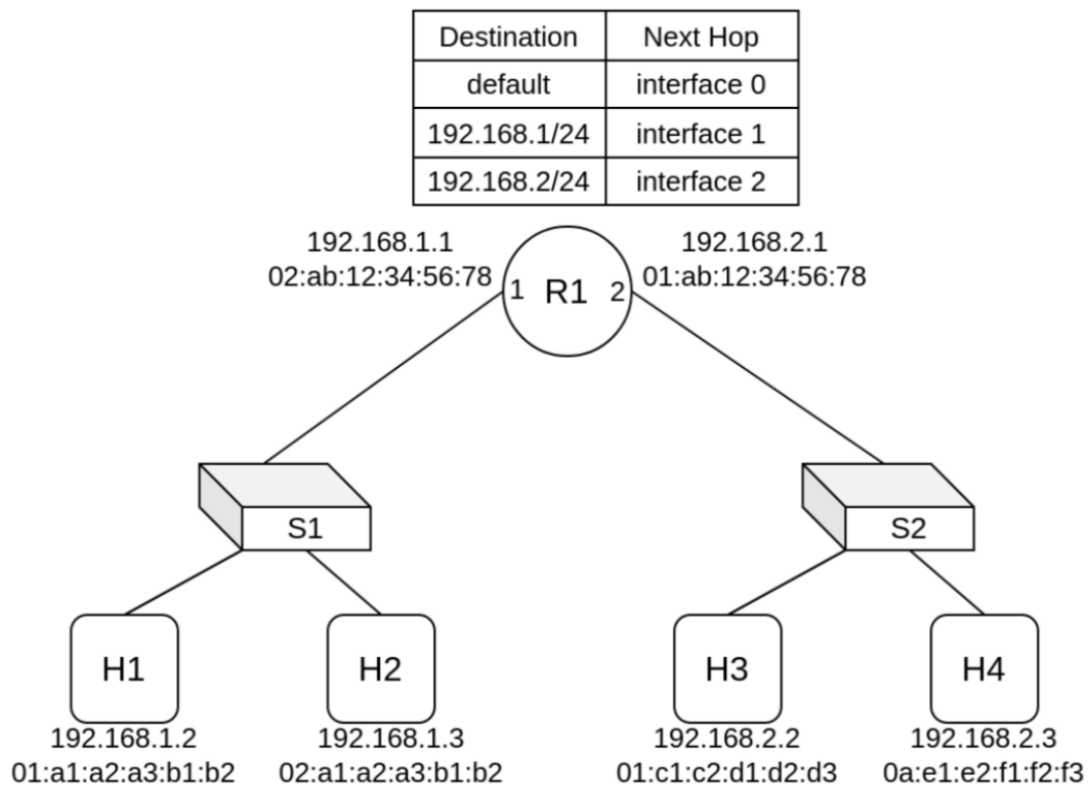
1. What is the purpose of ARP?
  - a) To map IP addresses to MAC addresses
  - b) To map MAC addresses to IP addresses
  - c) To map domain names to IP addresses
  - d) To map subnet masks to IP addresses
  
2. Pure (Un-slotted) ALOHA is:
  - a) Highly efficient
  - b) Highly centralized
  - c) Highly decentralized
  - d) Fully decentralized
  
3. In which network technology is CSMA/CD commonly used?
  - a) Ethernet
  - b) Wi-Fi
  - c) Fiber optics
  - d) Bluetooth

**True/False:**

- a. In networks using CSMA/CD, there is a chance of collision. [T / F]
- b. When an IP router between two Ethernet segments forwards an IP packet, it **does not** modify the destination IP address. [T / F]

### Question 2:

Consider the network topology shown in Figure 1 comprising router, switches, and hosts. The IP addresses and MAC addresses assigned to various interfaces are shown in the figure. Assume that the router's ARP table is empty at the start. **[Marks 10] [CLO 3]**



**Figure 1: Sample Network Topology**

Suppose that the following IP packet arrives at “interface 2” of router

| Ethernet Source   | Ethernet Destination | IP Source   | IP Destination |
|-------------------|----------------------|-------------|----------------|
| 0a:e1:e2:f1:f2:f3 | 01:ab:12:34:56:78    | 192.168.2.3 | 192.168.1.2    |

Router forwards the packet to interface 1.

- a) To successfully deliver the above IP packet, the router needs to send an ARP request. Which host(s) receive the ARP request? (Answer in the form e.g. H8, H9)

**[1 Mark]**

**b)** Mention the following header fields for the ARP request packet sent by the router.

**[4 Marks]**

| <b>Ethernet Source</b> | <b>Ethernet Destination</b> | <b>IP Source</b> | <b>IP Destination</b> |
|------------------------|-----------------------------|------------------|-----------------------|
|                        |                             |                  |                       |

**c)** Who replies to the ARP request?

**[1 Mark]**

**d)** After the above ARP request-reply exchange, mention the following header fields for the IP packet that gets delivered to the destination host.

**[4 Marks]**

| <b>Ethernet Source</b> | <b>Ethernet Destination</b> | <b>IP Source</b> | <b>IP Destination</b> |
|------------------------|-----------------------------|------------------|-----------------------|
|                        |                             |                  |                       |

## **SOLUTION QUIZ-06 BCS-6A**

### **QUESTION-01**

1. A
2. D
3. A

### **TRUE/FALSE**

1. True
2. True

### **QUESTION-02**

**a) Host H1 and H2**

**b)**

**Sol:**

| <b>Ethernet Source</b> | <b>Ethernet Destination</b> | <b>IP Source</b> | <b>IP Destination</b> |
|------------------------|-----------------------------|------------------|-----------------------|
| 02:ab:12:34:56:78      | ff:ff:ff:ff:ff:ff           | 192.168.1.1      | 192.168.1.2           |

**c) Sol: Host H1**

**d)**

**Sol:**

| <b>Ethernet Source</b> | <b>Ethernet Destination</b> | <b>IP Source</b> | <b>IP Destination</b> |
|------------------------|-----------------------------|------------------|-----------------------|
| 02:ab:12:34:56:78      | 01:a1:a2:a3:b1:b2           | 192.168.2.3      | 192.168.1.2           |

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|                                                                                   | <b>Program:</b>     | <b>BS (Software Engineering)</b> | <b>Semester:</b>    | <b>Spring 2024</b> |
|                                                                                   | <b>Duration:</b>    | <b>15 minutes</b>                | <b>Total Marks:</b> | <b>15</b>          |
|                                                                                   | <b>Paper Date:</b>  | <b>02-May-2024</b>               | <b>Section</b>      | <b>BSE-6A</b>      |
|                                                                                   | <b>Exam Type:</b>   | <b>Quiz 6 - Chapter 6</b>        | <b>Page(s):</b>     | <b>3</b>           |

**Student Name**

**Roll No.**

**Section:**

**Q1. Encircle the correct option:**

**[5 marks] [CLO 3]**

1. In a switch table, what information does the MAC address table store?

- a) IP addresses of connected devices
- b) Port and corresponding MAC addresses
- c) Domain Name System (DNS) entries
- d) Switch configuration settings

2. CRC is used for:

- a) Error detection
- b) Error correction
- c) error detection and correction
- d) none

3. In ALOHA, what happens if a collision occurs?

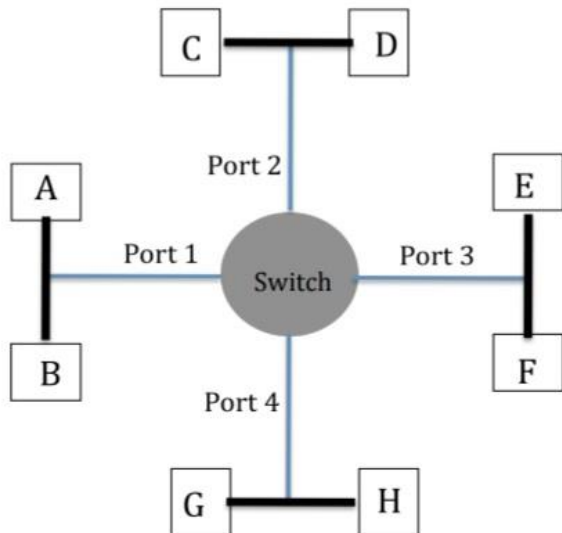
- a) The data is retransmitted after a random time
- b) The data is discarded
- c) The sender waits for an acknowledgment
- d) The sender reduces the transmission rate

**True/False:**

- i. Pure (Un-slotted) Aloha is highly centralized. [T / F]
- ii. In networks using CSMA/CD, there is a chance of collision [T / F]

### Question 2

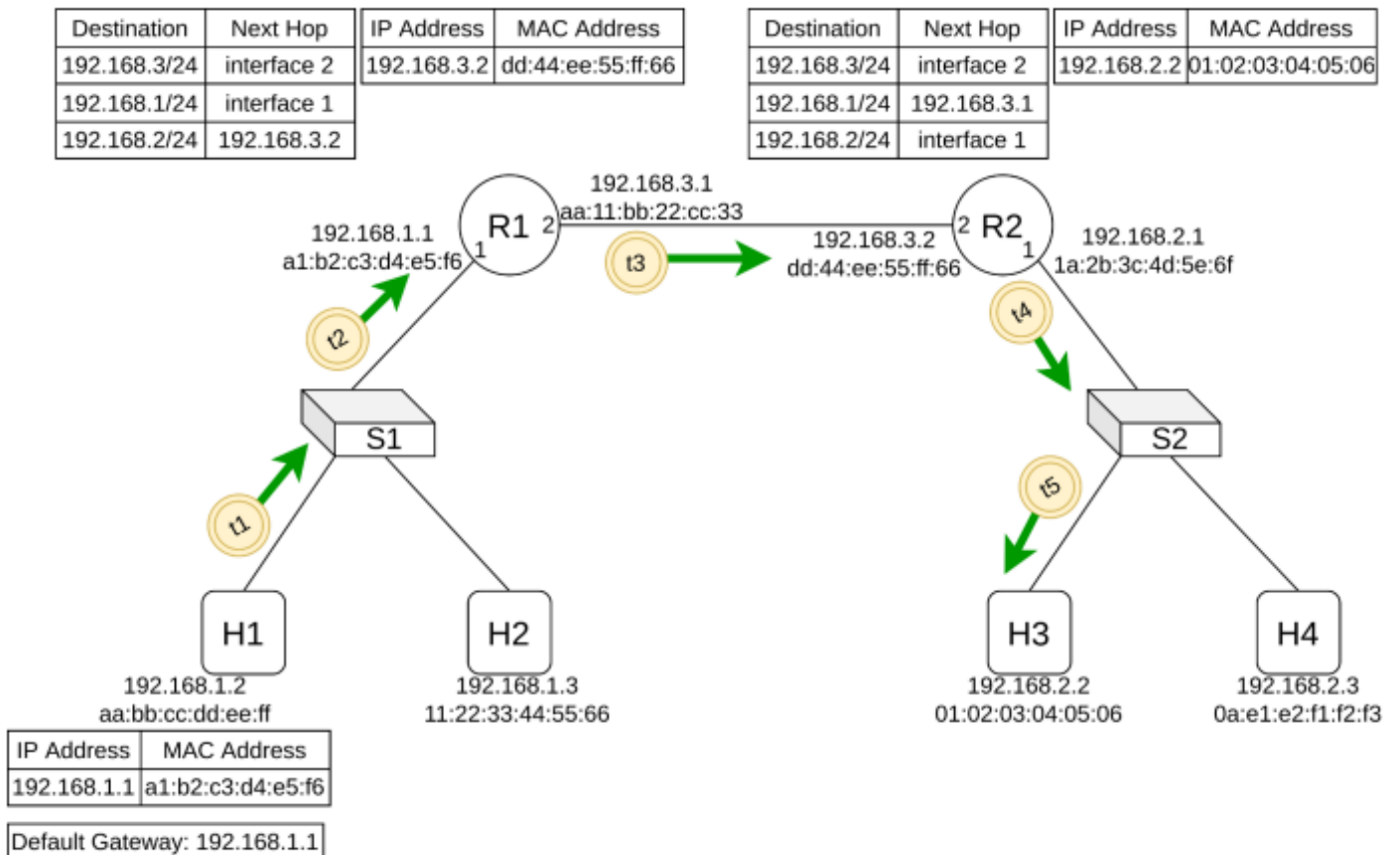
Consider the following set of **four Ethernet segments** connected by a central switch (not a hub). Each Ethernet segment has **two hosts** on it. Assume at the beginning that the switch has no state about where each of the hosts reside, and will use **self-learning** to build its **routing table** during the sequence of transmissions listed below. For each step in this sequence of transmissions, list which **port(s)** the Ethernet frame is forwarded to by the switch  
[1\*4 = 4 Marks] [CLO 3]



- i. A sends to E: \_\_\_\_\_
- ii. A sends to B: \_\_\_\_\_
- iii. G sends to E: \_\_\_\_\_
- iv. E sends to A: \_\_\_\_\_

### Question 3

Consider the network topology shown in Figure 2 comprising routers, switches, and hosts. The IP addresses and MAC addresses assigned to various interfaces are shown in the figure. The routing and ARP table entries are also shown for selected routers and hosts. The figure shows five timestamps in the journey of an IP packet from **source host 192.168.1.2 to destination host 192.168.2.2**.  
[3\*2=6 Marks] [CLO 3]



**Figure 2: Sample Network Topology**

Mention the following header fields for the above IP packet at timestamps **t1** to **t3**.

| Timestamp | Ethernet Source | Ethernet Destination | IP Source | IP Destination |
|-----------|-----------------|----------------------|-----------|----------------|
| t1        |                 |                      |           |                |
| t2        |                 |                      |           |                |
| t3        |                 |                      |           |                |

## **SOLUTION QUIZ-06 BSE-6A**

### **QUESTION-01**

- 1) B
- 2) A
- 3) A

### **TRUE/FALSE**

- i. False
- ii. True

### **QUESTION-02**

- i. A sends to E: 2,3,4
- ii. A sends to B: 2,3,4
- iii. G sends to E: 1,2,3
- iv. E sends to A: 1

### **QUESTION-03**

| <b>Timestamp</b> | <b>Ethernet Source</b> | <b>Ethernet Destination</b> | <b>IP Source</b> | <b>IP Destination</b> |
|------------------|------------------------|-----------------------------|------------------|-----------------------|
| <b>t1</b>        | aa:bb:cc:dd:ee:ff      | a1:b2:c3:d4:e5:f6           | 192.168.1.2      | 192.168.2.2           |
| <b>t2</b>        | aa:bb:cc:dd:ee:ff      | a1:b2:c3:d4:e5:f6           | 192.168.1.2      | 192.168.2.2           |
| <b>t3</b>        | aa:11:bb:22:cc:33      | dd:44:ee:55:ff:66           | 192.168.1.2      | 192.168.2.2           |

## SOLUTION QUIZ-05 BSE-6A

### QUESTION-01

#### MCQ's

1. d
2. d
3. d

#### T/F

1. F
2. T

### QUESTION-02

**Solution:** Refer to the binary division (XOR) operation below, the checksummed message sent is **1010101101 1111**

$$\begin{array}{r} 11001 \overline{) 10101011010000} \\ \underline{11001} \downarrow \\ 01100011010000 \\ \underline{11001} \downarrow \\ 0000111010000 \\ \underline{11001} \downarrow \\ 001000000 \\ \underline{11001} \downarrow \\ 0100100 \\ \underline{11001} \downarrow \\ 010110 \\ \underline{11001} \\ 01111| \end{array}$$

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|                                                                                   | <b>Program:</b>     | <b>BS (Software Engineering)</b> | <b>Semester:</b>    | <b>Spring 2025</b> |
|                                                                                   | <b>Duration:</b>    | <b>15 minutes</b>                | <b>Total Marks:</b> | <b>15</b>          |
|                                                                                   | <b>Paper Date:</b>  | <b>06-May-2025</b>               | <b>Section</b>      | <b>6A</b>          |
|                                                                                   | <b>Exam Type:</b>   | <b>Quiz 6 - Chapter 6</b>        | <b>Page(s):</b>     | <b>2</b>           |

**Student Name**

**Roll No.**

**Section:**

**Q1.**

**[5 marks] [CLO 3]**

**Pure (Un-slotted) ALOHA is:**

- a) Highly efficient
- b) Highly centralized
- c) Highly decentralized
- d) Fully decentralized

**What is the length of the preamble in the frame header?**

- a) 4 bytes
- b) 5 bytes
- c) 6 bytes
- d) 7 bytes

**It is possible to detect and correct multiple bit errors on the link layer using:**

- a) CRC
- b) Ethernet
- c) 2-D parity scheme
- d) None of the above

**True/False:**

The data link layer divides the stream of bits received from the network layer into data units called Datagram. **[T / F]**

Flow control can be implemented at the data link layer. **[T / F]**

**Q2.** Station A wants to send a data string  $D = 1010101101$  to station B using the CRC generator string  $G = 11001$  (equivalent to generator polynomial  $G(x) = x^4 + x^3 + 1$ ). What is the complete check-summed message that is sent? (Show complete working)[10 Marks] [CLO 3]

## **SOLUTION QUIZ-06 BSE-6B**

### **QUESTION-01**

1. B
2. C
3. D
4. C
5. C

### **QUESTION-02**

1. At time  $t=1$ , (H,7) is added as an entry to switch table 1.
2. At time  $t=1$ , (A,1) is added as an entry to switch table 1.
3. At time  $t=1$ , (H,11) is added as an entry to switch table 2.
4. At time  $t=1$ , (A,8) is added as an entry to switch table 2.
5. At time  $t=2$ , (K,7) is added as an entry to switch table 1.
6. At time  $t=2$ , switch table 1 doesn't observe this frame.
7. At time  $t=2$ , (K,14) is added as an entry to switch table 2
8. At time  $t=2$ , (I,12) is added as an entry to switch table 2.
9. None
10. None

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|                                                                                   | <b>Program:</b>     | <b>BS (Software Engineering)</b> | <b>Semester:</b>    | <b>Spring 2026</b> |
|                                                                                   | <b>Duration:</b>    | <b>20 minutes</b>                | <b>Total Marks:</b> | <b>15</b>          |
|                                                                                   | <b>Paper Date:</b>  | <b>05-May-2026</b>               | <b>Section</b>      | <b>6A , 6B</b>     |
|                                                                                   | <b>Exam Type:</b>   | <b>Quiz 6 - Chapter 6</b>        | <b>Page(s):</b>     | <b>2</b>           |

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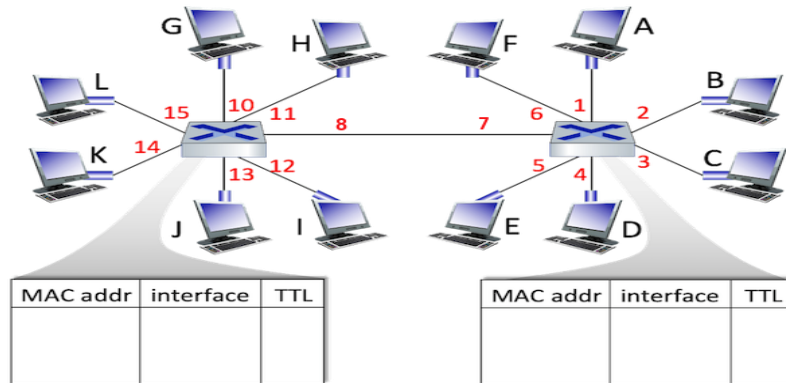
**Section:**

**Q1. Encircle the correct option:**

**[5 Marks] [CLO 3]**

- The DHCP Request message is encapsulated in which Transport Layer protocol?
  - TCP
  - UDP
  - ARP
  - It is not encapsulated in a Transport Layer; it sits directly on IP.
- After the DHCP handshake, the laptop sends a DNS query for www.google.com. The DNS Server is on a remote network. The frame containing this DNS query has which Destination MAC address?
  - The MAC address of the Google Web Server.
  - The MAC address of the DNS Server.
  - The MAC address of the First-Hop Router.
  - FF:FF:FF:FF:FF:FF
- A CRC system uses generator polynomial  $G(x) = 1101$ . Which of the following error patterns is guaranteed to be detected?
  - 1100
  - 1010
  - 1000
  - 111
- Which of the following generators satisfies both conditions: (1) reliably detects all single-bit errors, AND (2) reliably detects all odd-numbered bit errors?
  - 1000 ( $x^3$ )
  - 111 ( $x^2 + x + 1$ )
  - 1001 ( $x^3 + 1$ )
  - None
- A layer 2 switch can use:
  - flooding
  - selective forwarding
  - both a & b
  - none

**Q2.** Consider the LAN below consisting of 10 computers connected by two self-learning Ethernet switches. At  $t=0$  the switch table entries for both switches are empty. At  $t=1$  and  $2$ , a source node sends to a destination node as shown below, and the destination replies immediately (well before the next time step). **[10 Marks] [CLO 3]**



Assume that the following transmissions occur:

$t=1$ :  $H \rightarrow A$

$t=2$ :  $K \rightarrow I$

Please answer the following questions:

- At  $t=1$ , what is the source entry for switch 1?  
Answer: \_\_\_\_\_
- At  $t=1$ , what is the destination entry for switch 1?  
Answer: \_\_\_\_\_
- At  $t=1$ , what is the source entry for switch 2?  
Answer: \_\_\_\_\_
- At  $t=1$ , what is the destination entry for switch 2?  
Answer: \_\_\_\_\_
- At  $t=2$ , what is the source entry for switch  
Answer: \_\_\_\_\_
- At  $t=2$ , what is the destination entry for switch 1?  
Answer: \_\_\_\_\_
- At  $t=2$ , what is the source entry for switch 2?  
Answer: \_\_\_\_\_
- At  $t=2$ , what is the destination entry for switch 2?  
Answer: \_\_\_\_\_
- What is the physical address of interface 8 of the switch at the left side in the diagram?  
Answer: \_\_\_\_\_
- What is the physical address of interface 7 of the switch at the right side in the diagram?  
Answer: \_\_\_\_\_

# SOLUTION QUIZ-05 BSE-6C

## QUESTION-01

### MCQ's

1. a
2. c
3. d

### T/F

1. F
2. F

## QUESTION-02

### Part-A:

| <u>IP Address</u> | <u>Mac Address</u> | <u>Short Reason</u> |
|-------------------|--------------------|---------------------|
| IP-15             | MAC-C/MAC-15       | Same subnet         |
| IP-D/IP16         | MAC-D              | Same subnet         |

### Part-B:

| <u>IP Address</u> | <u>Mac Address</u> | <u>Correct Entry?(yes or no)</u> | <u>Short Reason</u>    |
|-------------------|--------------------|----------------------------------|------------------------|
| IP-A              | MAC-A              | No                               | Not on the same subnet |
| IP-20             | MAC-20             | No                               | Switch interface       |
| IP-E              | MAC-E              | Yes                              | Same subnet            |

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|                                                                                   |                     |                                  |                     |                    |
|-----------------------------------------------------------------------------------|---------------------|----------------------------------|---------------------|--------------------|
|  | <b>Course Name:</b> | <b>Computer Networks</b>         | <b>Course Code:</b> | <b>CS 3001</b>     |
|                                                                                   | <b>Program:</b>     | <b>BS (Software Engineering)</b> | <b>Semester:</b>    | <b>Spring 2025</b> |
|                                                                                   | <b>Duration:</b>    | <b>15 minutes</b>                | <b>Total Marks:</b> | <b>15</b>          |
|                                                                                   | <b>Paper Date:</b>  | <b>06-May-2025</b>               | <b>Section</b>      | <b>6C</b>          |
|                                                                                   | <b>Exam Type:</b>   | <b>Quiz 6 - Chapter 6</b>        | <b>Page(s):</b>     | <b>2</b>           |

**Student Name**

**Roll No.**

**Section:**

**Q1. Encircle the correct option:**

**[5 marks] [CLO 3]**

- Which of the following is the multiple access protocol for wired network channel?
  - CSMA/CD
  - CSMA/CA
  - Both
  - All
- The data link layer takes the packet from\_\_\_\_\_and encapsulates them into frame for transmission.
  - Segment
  - Frames
  - Datagram
  - Message
- It is possible to detect and correct multiple bit errors on link layer using:
  - CRC
  - Ethernet
  - 2 - D parity scheme
  - None of the above

**True/False:**

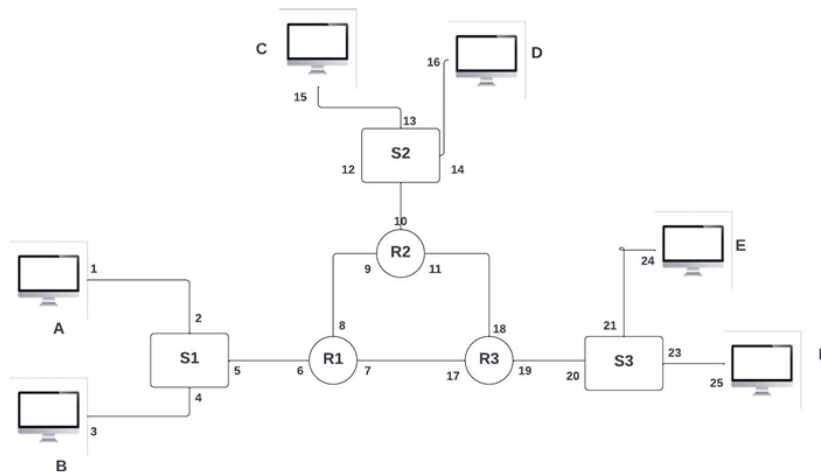
- Network Layer lies below the data link layer. [ T / F ]
- Minimum payload length in frame header is 56 bytes. [ T / F ]

**Question 2:**

**[Marks 10] [CLO 3]**

Consider the network scenario given in Figure comprising six end-systems/hosts: A, B, C, D, E and F with MAC addresses: MAC-A, MAC-B, MAC-C, MAC-D, MAC-E, MAC-F and IP addresses: IP-A, IP-B, IP-C, IP-D, IP-E, IP-F respectively. DHCP is enabled for the given network topology. S1, S2 and S3 denote the switches while routers are represented by R1, R2 and R3. Connecting lines depict the network connections (wired) between all the network nodes with individual digits on each end representing the interface number of the connected node.

Furthermore, for IP and MAC addresses of the interfaces, IP-(Interface number) and MAC- (Interface number)format is in use. For example, IP-6 and MAC-6 refers to the IP and MAC addresses of Router1's interface 6.



**Part (a):** Consider the following ARP table for interface 10 of the router R2 and complete the table with the help of entries given. For each entry, please give a brief reason for your choice. [4 Marks]

| <u>IP Address</u> | <u>Mac Address</u> | <u>Short Reason</u> |
|-------------------|--------------------|---------------------|
| IP-15             |                    |                     |
|                   | MAC-D              |                     |

**Part(b):** Consider the following ARP table for interface 19 of the router R3 and specify in the entries given are correct or not. For each entry, please give a brief reason for your choice. [6 Marks]

| <u>IP Address</u> | <u>Mac Address</u> | <u>Correct Entry?(yes or no)</u> | <u>Short Reason</u> |
|-------------------|--------------------|----------------------------------|---------------------|
| IP-A              | MAC-A              |                                  |                     |
| IP-20             | MAC-20             |                                  |                     |
| IP-E              | MAC-24             |                                  |                     |